

Unit 1

Introduction

Differential Equation

An equation which involves derivatives or differentials of one or more dependent variables with respect to one or more independent variables is called a differential equation.

Examples

1. $dy = (x + \sin x)dx$
2. $\frac{d^4x}{dt^4} + \frac{d^2x}{dt^2} + \left(\frac{dx}{dt}\right)^2 = e^2$
3. $y = \sqrt{x} \frac{dy}{dx} + \frac{k}{\frac{dy}{dx}}$
4. $k \left(\frac{d^2y}{dx^2}\right) = \left[1 + \frac{dy}{dx}\right]^{\frac{3}{2}}$
5. $\frac{\partial^2 v}{\partial t^2} = k \left(\frac{\partial^3 v}{\partial t^3}\right)^2$
6. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

Note

$$y' \text{ (or } y_1), y'' \text{ (or } y_2), \dots, y^{(n)} \text{ (or } y_n)$$

will denote

$$\frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n} \text{ respectively.}$$

Thus, equation (3) can be written as

$$y = \sqrt{x}y' + \frac{k}{y'} \quad \text{or,} \quad y = \sqrt{x} + \frac{k}{y_1}$$

Ordinary and Partial Differential Equations

Ordinary Differential Equation

An ordinary differential equation (ODE) is a differential equation which involves derivatives a single variable with respect to a single independent variable.

In the above examples, equations (1), (2), (3) and (4) are ordinary differential equations.

Partial Differential Equation (PDE)

A differential equation which involves partial derivatives of one or more dependent variables with respect to one or more independent variables is called a partial differential equation (PDE).

Examples

In the above examples, equations (5) and (6) are both partial differential equations.

Order of a Differential Equation

The order of the higher order derivative involved in a differential equation is called the order of the differential equation.

Examples

In the above examples, equations (1) and (3) are of first order, equation (2) is of fourth order, equations (4) and (6) are of the second order and equation (5) is of third order.

Degree of a Differential Equation

The degree of a differential equation is the degree of the highest order derivative present in it after the equation has been made free from the radicals or fractions.

Examples

In the above example equations (1), (2) and (6) are of first degree. Equation (3) after making free from fraction, becomes

$$y \frac{dy}{dx} = \sqrt{x} \left(\frac{dy}{dx} \right)^2 + k, \text{ is of second degree.}$$

Equation (4), which when squared both sides, becomes of degree 2 and equation (5) also is of degree 2.

Note

An ordinary differential equation of first order is generally denoted by

$$M(x, y)dx + N(x, y)dy = 0.$$

For example, $(y - x)dx + 4xdy = 0$.

In normal form a first order ordinary differential equation is of the form

$$\frac{dy}{dx} = f(x, y)$$

and on ODE of second order, in normal form, is of the form

$$\frac{d^2y}{dx^2} = f(x, y, y')$$

Symbolically, an n^{th} order ordinary differential equation in general form is

$$F(x, y, y', \dots, y^{(n)}) = 0.$$

In the normal form an n^{th} order ordinary differential equation is

$$\frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)}).$$

For example,

$$(i) \ 4xy' - y = x \Rightarrow y' = \frac{x - y}{4x}$$

$$(ii) \ y - y' + 6y = 0 \Rightarrow y' = y - 6y$$

Linear and Non-linear Differential Equations

A differential equation is said to be linear if (i) every dependent variable and every derivative present in the equation occur in the first degree only, and (ii) no products of dependent variables and (or) derivatives occur. A differential equation which is not linear is called non-linear.

Characteristic Properties of a Linear ODE

A linear ODE satisfies the following conditions/properties:

1. The dependent variable y and its derivatives $y', y'' \dots, y^{(n)}$ are of the first degree.
2. No product of y and/or any of its derivatives appear in the equation.
3. No transcendental function of y (such as $\sin y$, $\cos y$, $\tan y$, e^y or $\log y$ or $\cosh y$) and/or its derivatives occur.

Examples

1. $\frac{d^3y}{dx^3} + 5x \frac{d^2y}{dx^2} + y \sin x = 0$ (Third order linear)
2. $\frac{d^3y}{dx^3} + 5x \frac{d^2y}{dx^2} + y \sin y = 0$ (Non-linear since $\sin y$ is present)
3. $y' + e^y = x$ (First order non-linear, because of the presence of e^y)
4. $x^2y + xy' + y = \cos x$ (First order linear)
5. $y^{(4)} + (y')^2 + 7y = 0$ (non-linear because of $(y')^2$)
6. $(1 - y)(y'' + 2y) = e^x$ (non-linear)
7. $\cos xy'' + e^x y' = \sin x$ (Second order linear)
8. $\frac{d^2y}{dx^2} + y^2 \sin x = 0$ (non-linear)

$$9. \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \text{ (Second order linear)}$$

$$10. \frac{\partial^2 u}{\partial x^2} \cdot \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial x \partial y} \text{ (non-linear)}$$

Solution of Differential Equation

Any relation between the dependent and independent variables which when substituted in the differential equation reduces it to an identity is called a solution or an integral or a primitive of the differential equation.

In other words, a solution of a differential equation is a function that satisfies the differential equation identically in an interval I . The interval I is called the interval of definition (or interval of existence or interval of validity or the domain of solution).

A solution of an n^{th} order ODE

$$y(x) = f(x, y, y', \dots, y^{(n-1)})$$

is a function

$$y = f(x)$$

that possesses at least n derivatives i.e. y is n –times differentiable and for which

$$y^n(x) = f(x, y(x), y'(x), y''(x), \dots, y^{(n-1)}(x))$$

for all x in some interval I .

The process of solution of a differential equation is integration which brings with it an arbitrary constant whose possible values constitute an infinite family of solutions.

When we solve a first order differential equation $F(x, y, y') = 0$, we obtain a solution containing a single arbitrary constant c . A solution containing one arbitrary constant c is represented by

$$G(x, y, c) = 0$$

which is a family called one parameter family of solution.

Similarly, when we solve an n^{th} order differential equation $F(x, y, y', y'', \dots, y^{(n)}) = 0$ we obtain n -parameter family of solutions

$$G(x, y, c_1, c_2, \dots, c_n) = 0.$$

Solution of a System

The solution of a system of differential equation with two differential equations such as

$$\begin{aligned} \frac{dx}{dt} &= f(t, x, y) \\ \frac{dy}{dt} &= g(t, x, y) \end{aligned}$$

is a pair of differentiable functions

$$x = f_1(t) \text{ and } y = f_2(t)$$

defined in a common interval I which will satisfy the system in this interval.

Example 1 Show that $y = \left(\frac{A}{x}\right) + B$ is a solution of the differential equation $\frac{d^2 y}{dx^2} + \frac{2}{x} \times \frac{dy}{dx} = 0$.

Solution

The given ODE is

$$\frac{d^2 y}{dx^2} + \frac{2}{x} \times \frac{dy}{dx} = 0 \quad \dots \quad (1)$$

The given function is

$$y = \left(\frac{A}{x}\right) + B \quad \dots \quad (2)$$

Differentiating (2) w.r.t. x , we get

$$\frac{dy}{dx} = -\frac{A}{x^2}$$

Again, differentiating w.r.t. x , we get

$$\frac{d^2y}{dx^2} = \frac{2A}{x^3}$$

Now, substituting these values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in (2), we get

$$\begin{aligned}\frac{2A}{x^3} + \frac{2}{x} \times \left(-\frac{A}{x^2}\right) &= 0 \\ \Rightarrow \frac{2A}{x^3} - \frac{2A}{x^3} &= 0 \\ \Rightarrow 0 &= 0,\end{aligned}$$

which is true.

Therefore, $y = \left(\frac{A}{x}\right) + B$ is a solution of $\frac{d^2y}{dx^2} + \frac{2}{x} \times \frac{dy}{dx} = 0$.

Example 2 Show that $y = a \cos(mx + b)$ is a solution of the differential equation $\frac{d^2y}{dx^2} + m^2y = 0$.

Complete Primitive (General Solution), Particular Solution and Singular Solution

General Solution/Complete Primitive/Complete Integral

A solution of a differential equation containing arbitrary constants is called the general solution. The number of arbitrary constants in a general solution is equal to the order of the differential equation. That is, the general solution of an ODE of the first order contains one arbitrary constant, the general solution of an ODE of the second order contains two arbitrary constants and so on.

Particular Solution

A solution of a differential equation obtained from the general solution by assigning particular values to arbitrary constants is called a particular solution.

Singular Solution

A solution of a differential equation which can not be obtained from any general solution by assigning any particular values to the arbitrary constant is called a singular solution.

Example We can easily verify that $y = c_1 e^x + c_2 e^{2x}$... (1) is a general solution of the differential equation

$$y'' - 3y' + 2y = 0 \quad \dots \quad (2)$$

Since c_1 and c_2 are two arbitrary constants and the differential equation (2) is of order 2, (1) is a general solution of (2).

Some particular solutions of (2) are

$$y = e^x + e^{2x}, y = e^x - 2e^{2x} \text{ etc.}$$

Example 2 $y = (x + c)^2$ is a general solution of

$$\left(\frac{dy}{dx}\right)^2 - 4y = 0 \text{ (easily verified)} \quad \dots \quad (2)$$

We can also verify that $y = 0$ is a solution of (2). Moreover, $y = 0$ cannot be obtained by any choice of the arbitrary constant c in (1). Hence, $y = 0$ is a singular solution of (2).

Formation of Differential Equation

From a given family of curves, containing n arbitrary constants, we can obtain an n^{th} order differential equation, whose solution is the given family. For this, we follow the following steps:

1. Write down the equation of the given family of curves.
2. Differentiate this equation n times so as to get n additional equations containing the n arbitrary constants and derivatives.
3. Eliminate n arbitrary constants from the $n + 1$ equations obtained above.

Thus, we obtain an n^{th} order differential equation involving an n^{th} order derivative.

Examples

1. Find the differential equation of the family of curves $y = e^{mx}$ where m is an arbitrary constant.

Solution

The given curve is

$$y = e^{mx} \quad \dots \quad (1)$$

Differentiating (1) w.r.t. x , we get

$$y' = me^{mx} \quad \dots \quad (2)$$

Using (1) in (2), we get

$$\begin{aligned} y' &= my \\ \Rightarrow \frac{dy}{dx} &= my \\ \Rightarrow m &= \frac{1}{y} \frac{dy}{dx} \quad \dots \quad (3) \end{aligned}$$

Also, from (1), $mx = \ln y \Rightarrow m = \frac{\ln y}{x}$

From (3), $\frac{1}{y} \frac{dy}{dx} = \frac{\ln y}{x}$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} \ln y,$$

which is the required differential equation.

2. Find the differential equation which has $y = a \cos(mx + b)$ as its general solution, a and b are arbitrary constants and m be a given fixed constant.

Solution

The given curve is

$$y = a \cos(mx + b) \quad \dots \quad (1)$$

Differentiating (1) w.r.t. x , we get

$$\frac{dy}{dx} = -a m \sin(mx + b) \quad \dots \quad (2)$$

Again, differentiating w.r.t. x , we get

$$\frac{d^2y}{dx^2} = -am^2 \cos(mx + b) \quad \dots \quad (3)$$

$$\begin{aligned} \Rightarrow \frac{d^2y}{dx^2} &= -m^2 [a \cos(mx + b)] \\ \Rightarrow \frac{d^2y}{dx^2} &= -m^2 y, \text{ using (1)} \end{aligned}$$

The required differential equation is

$$\frac{d^2y}{dx^2} + m^2 y = 0$$

3. Form a differential equation from the equation of the curve $y = a \sin x + b \cos x + x \sin x$, where a and b are arbitrary constants.

$$\text{Ans: } \frac{d^2y}{dx^2} + y = 2 \cos x$$

Explicit and Implicit Solutions

A solution in which the dependent variable is expressed in terms of the independent variable is said to be Explicit solution. That is, an explicit solution is of the form

$$y = \phi(x).$$

For example

i. $y = \frac{x^4}{16}$ is explicit solution of $\frac{dy}{dx} = xy^{\frac{1}{2}}$.

ii. $y = \frac{1}{x}$ is an explicit solution of $xy' + y = 0$.

Implicit Solution

A solution of the form $f(x, y) = g(x, y)$ in which x and y are mixed together and y is not expressed in terms of x only.

For example, $y = x(x + y)^2$ is an example of implicit solution.

Initial Value Problems and Boundary Value Problems

Initial Value Problems

An ordinary differential equation together with some initial conditions is called an Initial Value Problem (IVP). Initial values tell where to start and an ODE tells where to go.

First order IVP

Solve $\frac{dy}{dx} = f(x, y)$ subject to $y(x_0) = y_0$

Second Order IVP

Solve $\frac{d^2y}{dx^2} = f(x, y, y')$ subject to $y(x_0) = y_0, y'(x_0) = y'_0$

An n^{th} Order Initial Value Problem is of the Form

Solve the n^{th} order ODE

$$\frac{d^n y}{dx^n} = f(x, y, y', y'', \dots, y^{(n-1)}),$$

subject to the n conditions,

$y(x_0) = y_0, y'(x_0) = y'_0, y''(x_0) = y''_0, \dots, y^{(n-1)}(x_0) = y_0^{(n-1)}$ is called an n^{th} order initial value problem.

The given conditions at the specified point x_0 are called the Initial/Auxiliary/Subsidiary Conditions.

Boundary Value Problems (BVP)

An ordinary differential equation together with Auxiliary Condition given at more than one value of x (the independent variable) is called a Boundary Value Problem (BVP). The Auxiliary Conditions in this case are called boundary conditions.

Examples

1. Solve $y'' + 2y' = e^x$ subject to $y(\pi) = 1$ and $y'(\pi) = 2$

2. Solve $y'' + 2y' = e^x$ subject to $y(0) = 1$ and $y'(0) = 2$

are initial value problems.

Examples

1. Solve $y'' + 2y' = e^x$ subject to $y(0) = 1$ and $y(3) = 2$ and

2. Solve $y'' + 2y' = e^x$ subject to $y(0) = y_a$ and $y(b) = y_b$ are the boundary value problems.

Mathematical Modeling

The main aim of application of mathematics is to translate the physical situation/real world problem, into mathematical equation. The process of translation of real life problems into mathematical formula is called mathematical modeling.

In order to apply differential equation for mathematical modeling, it is necessary to formulate an appropriate differential equation that properly describes/models the real world problem under investigation.

Most of the principles/laws that describe the behavior of the real world problems are statements/relations involving rates at which things happen. So, in order to understand and

investigate problems involving the motion of fluids, the flow of current in electric circuits, the dissipation of heat in solid objects, the propagation and direction of seismic waves, the increase or decrease of populations, it is necessary to know something about differential equations. The process of describing a real world physical situation with differential equation is called mathematical modeling. The differential equation which describes some physical situation is called the mathematical model.

Properties of a Good Mathematical Model

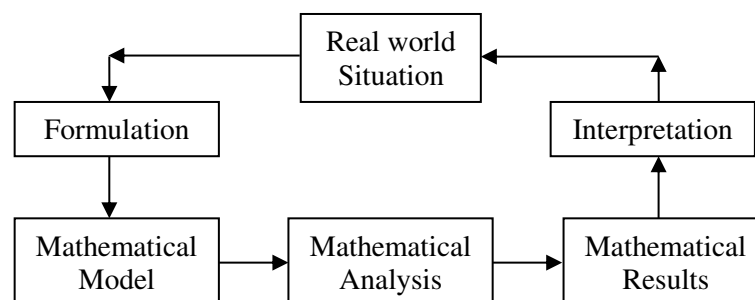
1. The Mathematical Model should be simple enough so that it can be solved easily.
2. The mathematical model should represent the actual situation sufficiently well so that the solution of the problem predicts the outcome of the real problem to within a useful degree of accuracy.

Construction of Mathematical Models

The following are the process for constructing mathematical models:

1. Identify the independent and dependent variables and parameters and use appropriate letters/symbols to represent them. For example, t is used to represent the independent variable time.
2. Choose appropriate units of measurement for each variable.
3. Clearly state the assumptions on which the model will be based on. (eg; Newton's law of motion)
4. Express the principle or law in step (3) in terms of the variables chosen.
5. Make sure that each term in the equation has the same physical units.
6. As the mathematical model is constructed, we need to compare prediction of the model with data from the system.

Mathematical Modeling



Modeling and Formation of Differential Equations

Example 1. Find the curves in xy –plane in the tangents at every point of which is perpendicular to the line joining the origin to that point.

Solution

A family of concentric circles with centre at origin.

Example 2. A particle moves in a straight line in such a way that its acceleration at a point is proportional to the square of its displacement measured from a fixed point on the line. Describe the motion mathematically.

Ans: $\frac{d^2y}{dx^2} = -\mu x^2$

Direction Field and Graphic Solution

The method of direction fields is a graphical method/way of displaying the general shape and behavior of the solutions of the ordinary first order differential equation $y' = f(x, y)$. The solution of first order ODE

$y' = f(x, y)$ is of the form $y = f(x)$ It is a fundamental method because it does not need to solve the differential equation $y' = f(x, y)$.

It was first discovered by L. Euler to draw a graphic to display the behavior of all solutions of the given ODE of first order without solving it.

Direction Field/Slope Field/Vector Field

The graph with many short/little tangent lines is known as a direction field.

An ODE of first order $y' = f(x, y)$ is replaced by a direction field model, a graphic consisting of pairs of grid points and line segments. A direction field line segment is also represented by an arrow to show the direction of the tangent. When drawing direction fields, we ask with ourselves the following questions:

- i. Where is the slope positive? negative? zero?
- ii. What is the slope when $x = 0$? , $y = 0$?
- iii. What is the slope when $y = x$?, when $y = -x$?
- iv. Does the slope depend on both x and y ? Just x ? Just y ?

Equilibrium Solution

The value of y obtained by putting $\frac{dy}{dx} = 0$ is called an equilibrium solution. The value separates the curves whose slopes are increasing (or decreasing) from those curves whose slopes are decreasing (or increasing).

Classification of Equilibrium Solution

1. If the solution curves are increasing below and decreasing above the equilibrium solution, the equilibrium solution is called stable or asymptotically stable.
2. If the solution curves are decreasing below and increasing above equilibrium solution, then the equilibrium solution is unstable.
3. Solution curves decreasing below and above or increasing below and above, the equilibrium solution is semi-stable.

Example

Draw a direction field for the differential equation

$y' = 3 - 2y$. Determine the behavior of y as $t \rightarrow \infty$. If this behaviour depends on the initial values of y at $t = 0$, describe the dependence.

Solution

The given differential equation is

$$y' = 3 - 2y \quad \dots \quad (1)$$

For equilibrium solution, let $y' = 0$. Therefore,

$$3 - 2y = 0 \Rightarrow y = \frac{3}{2} = 1.5$$

The equilibrium solution is

$$y = 1.5$$

To find the slope of the solution curves of (1), we put different values of y in (1).

For example,

Put $y = 2, y' = -1$

$$y = 3, y' = -3$$

$$y = 4, y' = -5$$

... and so on.

And put $y = 1, y' = 1$

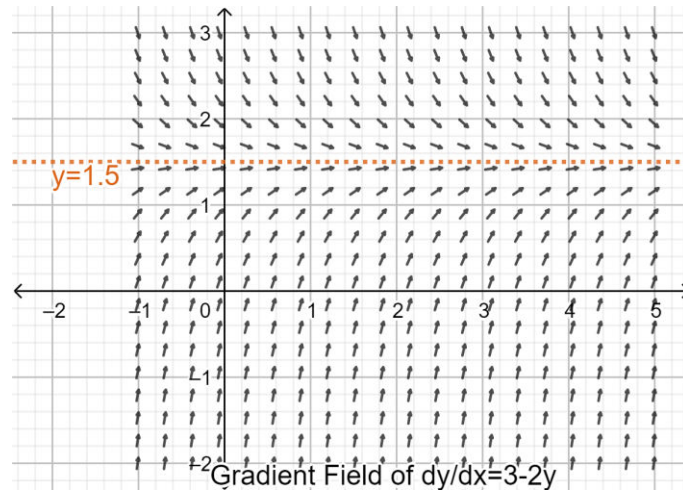
$y = 0, y' = 3$

$y = -1, y' = 5$

... and so on.

From this we see that $y' < 0$ for $y > \frac{3}{2}$ and the solution curves of (1) are decreasing. And for $y < \frac{3}{2}, y' > 0$ and the solution curves are increasing.

The direction field of (1) is as shown below.



We see from the figure that the direction field converts to $y = \frac{3}{2}$ as $t \rightarrow \infty$.

Example Investigate the behavior of solutions of the equation $\frac{dv}{dt} = 9.8 - \frac{v}{5}$, without solving it.

Solution

The given differential equation is

$$\frac{dv}{dt} = 9.8 - \frac{v}{5} \quad \dots \quad (1)$$

For equilibrium solution we put $\frac{dv}{dt} = 0$ in (1) to get

$$9.8 - \frac{v}{5} = 0 \Rightarrow \frac{v}{5} = 9.8 \Rightarrow v = 49.$$

The equilibrium solution is

$$v = 49.$$

To study the behavior of the solutions of (1), we find the slopes of the tangents at different points of tv -plane.

If we put $v = 40$, we get $\frac{dv}{dt} = 1.8$. That is, the slope of a solution $v = v(t)$ is 1.8 at any point where $v = 40$

Similarly,

$$\text{if } v = 45, \frac{dv}{dt} = 0.8$$

$$v = 35, \frac{dv}{dt} = 2.8$$

$$v = 30, \frac{dv}{dt} = 3.8$$

Again,

$$\text{if } v = 50, \frac{dv}{dt} = -0.2$$

$$v = 55, \frac{dv}{dt} = -1.2$$

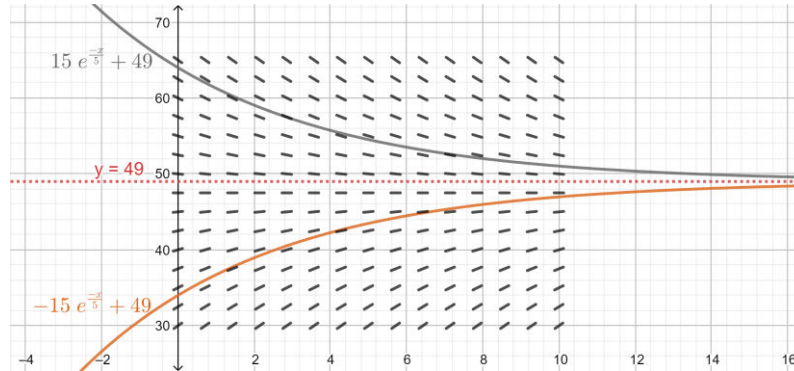
$$v = 60, \frac{dv}{dt} = -2.2$$

$$v = 65, \frac{dv}{dt} = -3.2$$

and so on.

We now display this information graphically in the tv -plane by drawing short line segment (tangent lines) with slope 1.8 at several points on the line $v = 40$, with slope $\frac{dv}{dt} = 0.8$ at several points on the line $v = 45$ and so on.

From this we see that $\frac{dv}{dt} > 0$ if $v < 49$ and $\frac{dv}{dt} < 0$ if $v > 49$.



Direction Field with Equilibrium Solution

Also, we see that solution curves above equilibrium solution are decreasing and below it, are increasing. So, the solutions of the equation (1) are converging to the equilibrium solution $v = 49$ as $t \rightarrow \infty$. That is, $v = 49$ is stable solution.

Example Draw a direction field for the differential equation. Based on the direction field, determine the behavior of y as $t \rightarrow \infty$. If this behavior depends on the initial value of y , describe the dependence:

$$y' = (y + 1)(y - 2)$$

Solution

The given differential equation is

$$\frac{dy}{dt} = (y + 1)(y - 2) \quad \dots \quad (1)$$

For equilibrium solution, $\frac{dy}{dt} = 0$

$$\Rightarrow (y + 1)(y - 2) = 0$$

$$\therefore y = -1, y = 2$$

The equilibrium solutions are

$$y = -1 \text{ and } y = 2.$$

To investigate the behavior of the solution of (1), we find the slopes of tangents at different points of ty -plane.

Put $y = -2, \frac{dy}{dt} = 4$ and so on.

So, if $y < -1, \frac{dy}{dt} > 0$

That is, the solution curves are increasing for $y < -1$ i.e. below $y = -1$.

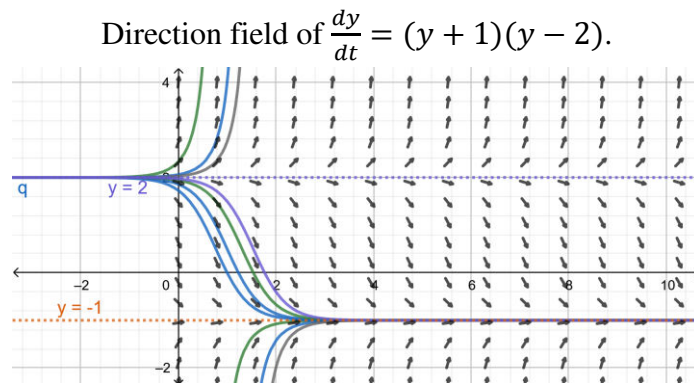
Putting $y = 0$, we get $\frac{dy}{dt} = -2 < 0$.

That is, for $-1 < y < 2, \frac{dy}{dt} < 0$, the solution curves are decreasing, in between $y = 2$ and $y = -1$.

Next, if $y = 3, \frac{dy}{dt} = 4 > 0$.

That is, for $y > 2, \frac{dy}{dt} > 0$ i.e. the solution curves are increasing above $y = 2$.

Since the solution curves above $y = 2$ are increasing and decreasing below it, the solution $y = 2$ is unstable. Again, since the solution curves above $y = -1$ are decreasing and below it are increasing, the solution $y = 1$ is asymptotically stable.



Example For the differential equation $\frac{dy}{dt} = y(y - 3)$, find the equilibrium solutions and sketch at least three possible solution curves for different values of $y(0) = y_0$ one in each region.

Solution

The given differential equation is

$$\frac{dy}{dt} = y(y - 3) \quad \dots \quad (1)$$

For equilibrium solution, we put $\frac{dy}{dt} = 0$. Then, (1) implies

$$\begin{aligned} y(y - 3) &= 0 \\ \Rightarrow y &= 0 \text{ and } y = 3 \end{aligned}$$

Therefore, the equilibrium solutions are

$$y = 0 \text{ and } y = 3.$$

We find the slopes of tangents to the solution curves at different points of ty -plane to describe the behavior of the solutions of (1).

Putting $y = -1, \frac{dy}{dt} = 4$

$y = -2, \frac{dy}{dt} = 10$, and so on.

Therefore, $\frac{dy}{dt} > 0$ for $y < 0$.

That is, the solution curves are increasing for $y < 0$.

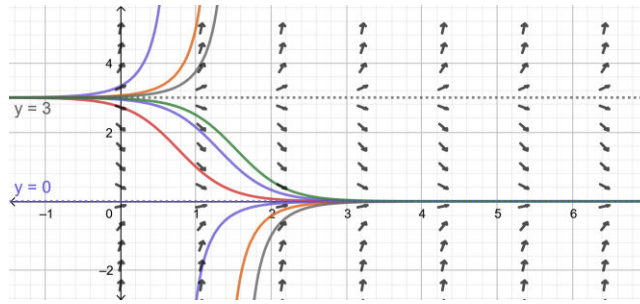
Next, putting $y = 1, \frac{dy}{dt} = -2 < 0$

$$y = 2, \frac{dy}{dt} = -2$$

Therefore, $\frac{dy}{dt} < 0$ for $0 < y < 3$. That is, the solution curves between 0 and 3 are decreasing.

Again, putting $y > 3$, we get $\frac{dy}{dt} > 0$. Thus, the solution curves are increasing for $y > 3$.

Direction field for $y' = y(y - 3)$



The equilibrium solution $y = 0$ is asymptotically stable and $y = 5$ is unstable.

Example Solve the following initial value problem and plot the solutions for several values of y_0 :

$$\frac{dy}{dt} = -2y + 10; \quad y(0) = y_0$$

Solution

The given IVP is

$$\frac{dy}{dt} = -2y + 10; \quad y(0) = y_0 \quad \dots \quad (1)$$

Now, $\frac{dy}{dt} = -2y + 10$

$$\begin{aligned} \Rightarrow \frac{dy}{-2(y-5)} &= dt \\ \Rightarrow \frac{dy}{(y-5)} &= -2dt \end{aligned}$$

Integrating, we get

$$\begin{aligned} \ln(y-5) &= -2t + \ln c \\ \Rightarrow \ln \frac{y-5}{c} &= -2t \\ \Rightarrow \frac{y-5}{c} &= e^{-2t} \\ \Rightarrow y-5 &= ce^{-2t} \\ \Rightarrow y &= ce^{-2t} + 5 \end{aligned}$$

The general solution of (1) is

$$y = ce^{-2t} + 5.$$

Using $y(0) = y_0$, we get

$$y_0 = c + 5 \Rightarrow c = y_0 - 5$$

The required solution of the IVP (1) is

$$y(t) = (y_0 - 5)e^{-2t} + 5 \quad \dots \quad (2)$$

Also, for equilibrium solution, we put $\frac{dy}{dt} = 0$.

Then,

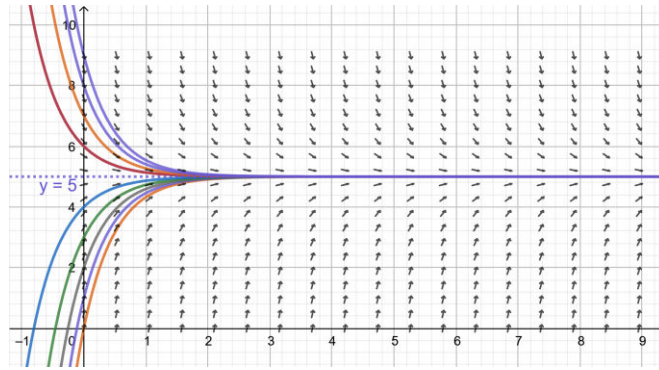
$$-2y + 10 = 0 \Rightarrow y = 5.$$

The equilibrium solution is

$$y = 5.$$

Graph of $y(t) = (y_0 - 5)e^{-2t} + 5$

For $y = 0, 1, 2, 3, 4, 6, 7, 8, 9$



Example The following object satisfies the initial value problem

$$\frac{dv}{dt} = 9.8 - \frac{v}{5}, \quad v(0) = 0$$

- Find the time that must elapse for the object to reach 98% of its limiting velocity.
- How far does the object fall in the time found in part (a)?

Solution

The given differential equation is (IVP)

$$\frac{dv}{dt} = 9.8 - \frac{v}{5}, \quad v(0) = 0 \quad \dots \quad (1)$$

Now, for equilibrium solution, $\frac{dv}{dt} = 0$

$$\begin{aligned} &\Rightarrow 9.8 - \frac{v}{5} \\ &\Rightarrow v = 49 \end{aligned}$$

The equilibrium solution is

$$v = 49 \quad \dots \quad (2)$$

Next, to find the general solution of (1)

$$\begin{aligned} \frac{dv}{dt} &= 9.8 - \frac{v}{5} \\ \text{or, } \frac{dv}{dt} &= \frac{49 - v}{5} \\ \text{or, } \frac{dv}{49 - v} &= \frac{dt}{5} \end{aligned}$$

Integration $\Rightarrow -\ln(49 - v) = \frac{1}{5}t + \ln c$

$$\Rightarrow \ln c - \ln(49 - v) = \frac{1}{5}t$$

$$\Rightarrow \ln \frac{c}{49 - v} = \frac{1}{5}t$$

$$\Rightarrow \frac{c}{49 - v} = e^{\frac{t}{5}}$$

$$\Rightarrow \frac{49 - v}{c} = e^{-\frac{t}{5}}$$

$$\Rightarrow 49 - v = ce^{-\frac{t}{5}}$$

$$\Rightarrow v = 49 - ce^{-\frac{t}{5}}$$

Therefore, the general solution of $\frac{dv}{dt} = 9.8 - \frac{v}{5}$ is

$$v(t) = 49 - ce^{-\frac{t}{5}} \quad \dots \quad (3)$$

Using initial condition $v(0) = 0$, we get

$$0 = 49 - ce^0 \Rightarrow c = 49.$$

The required solution of the IVP (1) is

$$v(t) = 49 - 49e^{-\frac{t}{5}}$$

or, $v(t) = 49 \left(1 - e^{-\frac{t}{5}}\right) \dots (4)$

a.

$$\begin{aligned} 98\% \text{ of } v(t) &= 49 \left(1 - e^{-\frac{t}{5}}\right) \times \frac{98}{100} \\ &= 49 \left(1 - e^{-\frac{t}{5}}\right) \times \frac{49}{50} \\ &= \frac{49^2}{50} \left(1 - e^{-\frac{t}{5}}\right) \end{aligned}$$

As $t \rightarrow \infty$,

$$98\% \text{ of } v(t) = \frac{49^2}{50}$$

Let T be the time required to reach this velocity. Then,

$$\begin{aligned} \frac{49^2}{50} &= 49 \left(1 - e^{-\frac{T}{5}}\right) \\ \Rightarrow \frac{49}{50} &= \left(1 - e^{-\frac{T}{5}}\right) \\ \Rightarrow e^{-\frac{T}{5}} &= 1 - \frac{49}{50} \\ \Rightarrow e^{-\frac{T}{5}} &= \frac{1}{50} \Rightarrow e^{\frac{T}{5}} = 50 \\ \Rightarrow \frac{T}{5} &= \ln 50 \Rightarrow T = 5 \ln 50 \end{aligned}$$

Therefore, $T = 19.56 \text{ sec}$.

Example Verify that the given function is the solution of the differential equation:

$$t^2 y'' + 5ty' + 4y = 0, \quad t > 0$$

$$(i) y_1(t) = \frac{1}{t^2} \quad (ii) y_2(t) = \frac{1}{t^2} \ln t$$

Solution

The given differential equation is

$$t^2 y'' + 5ty' + 4y = 0 \quad \dots (1)$$

The general solution is

$$y_2(t) = \frac{1}{t^2} \ln t \quad \dots (2)$$

Differentiating (2) w.r.t t , we get

$$y_2'(t) = \frac{t^2 \times \frac{1}{t} - 2t \ln t}{t^4} = \frac{t - 2t \ln t}{t^4} = \frac{1 - 2 \ln t}{t^3}$$

Again, differentiating w.r.t t , we get

$$\begin{aligned} y_2''(t) &= \frac{t^3 \times \frac{-2}{t} - 3t^2(1 - 2 \ln t)}{t^6} \\ &= \frac{-2t^2 - 3t^2(1 - 2 \ln t)}{t^6} \end{aligned}$$

$$= \frac{-2 - 3(1 - 2\ln t)}{t^4}$$

$$= \frac{-5 + 6\ln t}{t^4}$$

Putting these values in (1), we get

$$t^2 y'' + 5ty' + 4y = 0$$

$$\Rightarrow t^2 \times \frac{-5 + 6\ln t}{t^4} + 5t \times \frac{1 - 2\ln t}{t^3} + 4 \times \frac{1}{t^2} \ln t = 0$$

$$\Rightarrow \frac{-5 + 6\ln t}{t^2} + \frac{5 - 10\ln t}{t^2} + \frac{4\ln t}{t^2} = 0$$

$$\Rightarrow -5 + 6\ln t + 5 - 10\ln t + 4\ln t = 0 \Rightarrow 0 = 0, \text{ true.}$$

Hence, $y_2(t) = \frac{1}{t^2} \ln t$ is a solution of the differential equation (1).

Example Initially, a biologist begins the culture of new species of bacteria by putting 100 individuals. After 6 hrs, he finds 450 bacteria. If the growth of population of bacteria is proportional to the population at that time with a growth constant k , find the growth constant k .

Solution

Let $P(t)$ be the population of bacteria in time t (in hrs).

Then, by question,

$$\text{at } t = 0, p(0) = 100$$

$$\text{and at } t = 6, p(6) = 450.$$

Also, the rate of grow of bacteria is given by

$$\frac{dP(t)}{dt} = kP(t) \quad \dots \quad (1)$$

where, k is growth constant.

Now,

$$\frac{dP}{dt} = kP \Rightarrow \frac{dP}{P} = kdt \Rightarrow \ln P = kt + \ln c$$

$$\Rightarrow \ln \frac{P}{c} = kt \Rightarrow \frac{P}{c} = e^{kt} \Rightarrow P = ce^{kt}$$

Therefore, the general solution is

$$P = ce^{kt}.$$

Given that $P(0) = 100$, we have

$$100 = ce^0 \Rightarrow c = 100.$$

The particular solution is

$$P(t) = 100e^{kt}$$

Next, to find k , we use $P(6) = 450$, to get

$$450 = 100e^{6k} \Rightarrow e^{6k} = 4.5$$

$$\Rightarrow 6k = \ln 4.5 \Rightarrow k = \frac{\ln 4.5}{6} \Rightarrow k = 0.25$$

Hence, $k = 0.25$.